002_rapport_Messelu_JIN

2024-11-18

```r
set.seed(123)
a = 0.2
mu_1 = 0
mu_2 =1
s_1 = 3
s_2 = 1
Z = 1-a

#copy and paste pdf,CDF, inverse_F that were defined in the first project

f <- function(a, mu_1, mu_2, s_1, s_2, x) {
  C = 1 / (1 - a)
  f1 = (1 / (2 * pi * (s_1^2))^0.5) * exp(-0.5 * ((x - mu_1)^2) / (s_1^2))
  f2 = (1 / (2 * pi * (s_2^2))^0.5) * exp(-0.5 * ((x - mu_2)^2) / (s_2^2))
  return(C * (f1 - a * f2))
}

F <- function(a, mu_1, mu_2, s_1, s_2, x) {
  F1 <- pnorm(x, mean = mu_1, sd = s_1)
  F2 <- pnorm(x, mean = mu_2, sd = s_2)
  return((F1 - a * F2) / (1 - a))
}


inverse_F<-function(u,a, mu_1, mu_2, s_1, s_2,eps)
{
  if (u>=1 || u<=0) {stop("Error message: the value of CDF should be in ]0,1[")}

  x1 = -1 #initialize two values of x
  x2 = 1
  i=1

  #The while loop expands the bounds (x1 and x2) until F(x1) < u < F(x2),
  while((F(a, mu_1, mu_2, s_1, s_2, x1)-u)>0 || (F(a, mu_1, mu_2, s_1, s_2, x2)-u)<0){
    if(F(a, mu_1, mu_2, s_1, s_2, x1)-u>0){
       x1 <- x1*(2^i) # Expand x1 in the negative direction
    }
    if(F(a, mu_1, mu_2, s_1, s_2, x2)-u<0){
       x2 <- x2*(2^i)  # Expand x2 in the positive direction
    }
    i <- i+1
  }

  x = x1
```

```
  while(abs(x1-x2)>eps){
    x = (x1+x2)/2  # Calculate midpoint
    if((F(a, mu_1, mu_2, s_1, s_2, x)-u)<0){
      x1 <-x # Narrow down the interval
    }else{
      x2<-x
    }
  }
  return(x)

}
```

Question 9: Describe an accept-reject algorithm using g and Partition

To implement the accept-reject algorithm, we first need to identify a candidate density, denoted as h. This density h is obtained by normalizing g.

Let $h = Cg$ where $C \in \mathbb{R}$ such that h is a density.

$$C \int_{\mathbb{R}} g(x)dx = \frac{C}{Z}\left(\int_{D_0} f_1(x)dx + \sum_{i=1}^{k}\left(\sup_{D_i} f_i - a \inf_{D_i} f_2\right)\lambda(D_i)\right) = 1,$$

where $Z = 1 - a$ and λ is the Lebesgue measure.

Thus:

$$C = \frac{Z}{\int_{D_0} f_1(x)dx + \sum_{i=1}^{k}\left(\sup_{D_i} f_i - a \inf_{D_i} f_2\right)\lambda(D_i)}.$$

We also denote Ig the integral g. We have $Ig = \frac{1}{C}$

Then we can take the least upper bound $\widetilde{M}$ of $\frac{f}{g}$ which is:

$$\widetilde{M} = \max\left(\sup_{D_0}(1 - a\frac{f_2}{f_1}), \sup_{D_1}\frac{f_1 - af_2}{\sup_{D_1} f_1 - a\inf_{D_1} f_2}, ..., \sup_{D_k}\frac{f_1 - af_2}{\sup_{D_k} f_1 - a\inf_{D_k} f_2})\right) = 1.$$

Now we take M as the least upper bound of $\frac{f}{h}$. We have

$$M = \sup\frac{f}{h} = \sup\frac{f}{Cg} = \frac{1}{C}\sup\frac{f}{g} = \frac{1}{C}\widetilde{M} = \frac{1}{C}$$

because $\widetilde{M}$ is equal to 1.

Algorithm Steps

1. Simulate $Y \sim h$, $U \sim \mathcal{U}[0, 1]$.
2. Accept $X = Y$ if:
$$U \leq \frac{f(Y)}{Mh(Y)} = \frac{f(Y)}{g(Y)}.$$
3. Otherwise, return to step 1.

The theoretical acceptance rate is:

$$\frac{1}{M} = C = \frac{1}{Ig}.$$

Question 10: prove the existence of Partition

Let $\delta \in [0, 1[$. Define P_n:

$$D_0 =]-\infty, -n[\cup]n, +\infty[,$$

and

$$D_k = \left[-n + \frac{k-1}{n}, -n + \frac{k}{n}\right] \quad \text{for } k = 1, \ldots, 2n^2.$$

When $n \to \infty$,

$$\int_{D_0} f_1(x)dx \to 0$$

$$\sum_{i=1}^{k} \left(\sup_{D_i} f_i - a \inf_{D_i} f_2 \right) \lambda(D_i)$$

is Riemman sum whose limit converges to

$\int_{\mathbb{R}} f_1 - a \int_{\mathbb{R}} f_2$

we have the acceptance rate:

$$C \to \frac{1-a}{\int_{\mathbb{R}} f_1 - a \int_{\mathbb{R}} f_2} = \frac{1-a}{1-a} = 1.$$

Therefore, there exists $N \in \mathbb{N}$ such that:

$$\forall n \geq N, |C - 1| < \varepsilon.$$

We take $\varepsilon = 1 - \delta$. Hence:

$$C > \delta.$$

Thus:

$$\frac{1}{M} > \delta.$$

Conclusion We have shown the existence of $N \in \mathbb{N}$ that depends on δ such that for any $n \geq N$, the acceptance rate of P_n is greater than δ.

Question 11: stratified(n)

```r
#define a Partition of R by using a list of a, b, k to represent: Do = ]-inf,a]U]b,inf[, Di=]a+(b-a)/k*
P<-function(a,b,k)
{
   return(list(a,b,as.integer(k)))
}

#define a function taking extremes of interval and mean of Gaussian density, returning a maximizer of t
inter_argmax<-function(a1,b1,mu){
  if (mu <= a1){
    return (a1)
  }else if (mu>=b1){
    return (b1)
  }else {
    return (mu)
  }
}
#define a function taking extremes of interval and mean of Gaussian density, returning a minimizer of t
inter_argmin<-function(a1,b1,mu){
  if (abs(a1-mu)>=abs(b1-mu)){
    return (a1)
  }else{
    return (b1)
  }
}


#define a function taking partition p as argument to calculate Integral of g
Integral_g<-function(p)
{
  b = p[[1]]
  c = p[[2]]
  k = p[[3]]
  res = 0
  partition_points = seq(b,c,length.out = k+1)
  linear_heights = numeric(k+1)
  linear_heights[1] = pnorm(b,mu_1,s_1)
  l = (c-b)/k

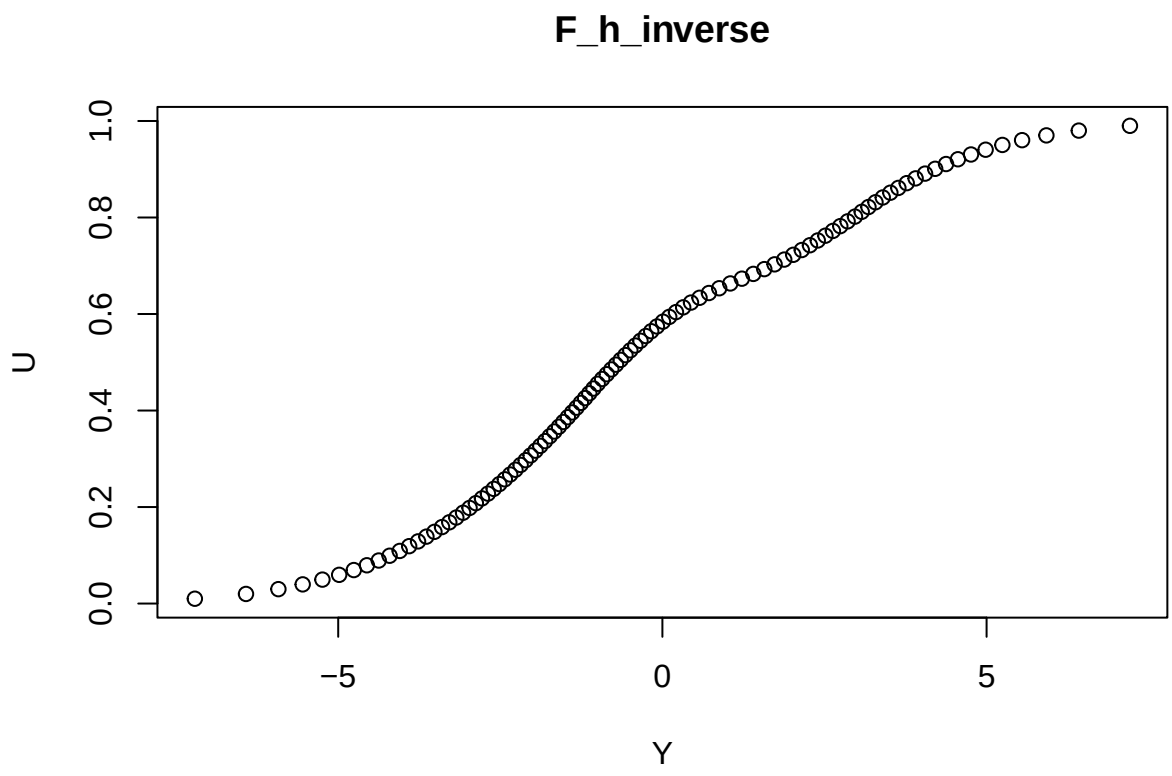
  for (i in (1:k)){
    a1 = b + l*(i-1)
    b1 = b + l*i
    sup = dnorm(inter_argmax(a1,b1,mu_1),mu_1,s_1)
    inf = dnorm(inter_argmin(a1,b1,mu_2),mu_2,s_2)
    linear_heights[i+1] = linear_heights[i] + (sup - a*inf)*l
  }
  res = (linear_heights[k+1]+1-pnorm(c,mu_1,s_1))
  return (list(Ig = res/Z,pp = partition_points, lh = linear_heights/res))
}

#define a Partition to be used in the function stratified to achieve acceptance rate bigger than 0.9
P0 = P(-1,3,160) # provide a partition
res_Ig = Integral_g(P0)
Ig = res_Ig$Ig #Intergral of g
```

```r
pp = res_Ig$pp #Partition points
lh = res_Ig$lh #Linear points of F_h_inverse on the extremes of intervals of the partition


#define inverse CDF of h
F_h_inverse<-function(p,u){
  b = p[[1]]
  c = p[[2]]
  k = p[[3]]
  if (u<=lh[1]){
    return (qnorm(u*Z*Ig,mu_1,s_1))
  }else if (u>=lh[k+1]){
    return(qnorm(1-Z*Ig*(1-u),mu_1,s_1))
  }else{
    i = findInterval(u,lh)
    return ((u-lh[i])*(pp[i+1]-pp[i])/(lh[i+1]-lh[i])+pp[i]) #On each interval Di, i=1,..,k, inverse CD
    }
}

#graphically check that F_h_inverse is correct
U=seq(0.01,0.99,length.out=100)
Y=sapply(U,function(i) F_h_inverse(P0,i))
plot(Y,U,main="F_h_inverse")
```

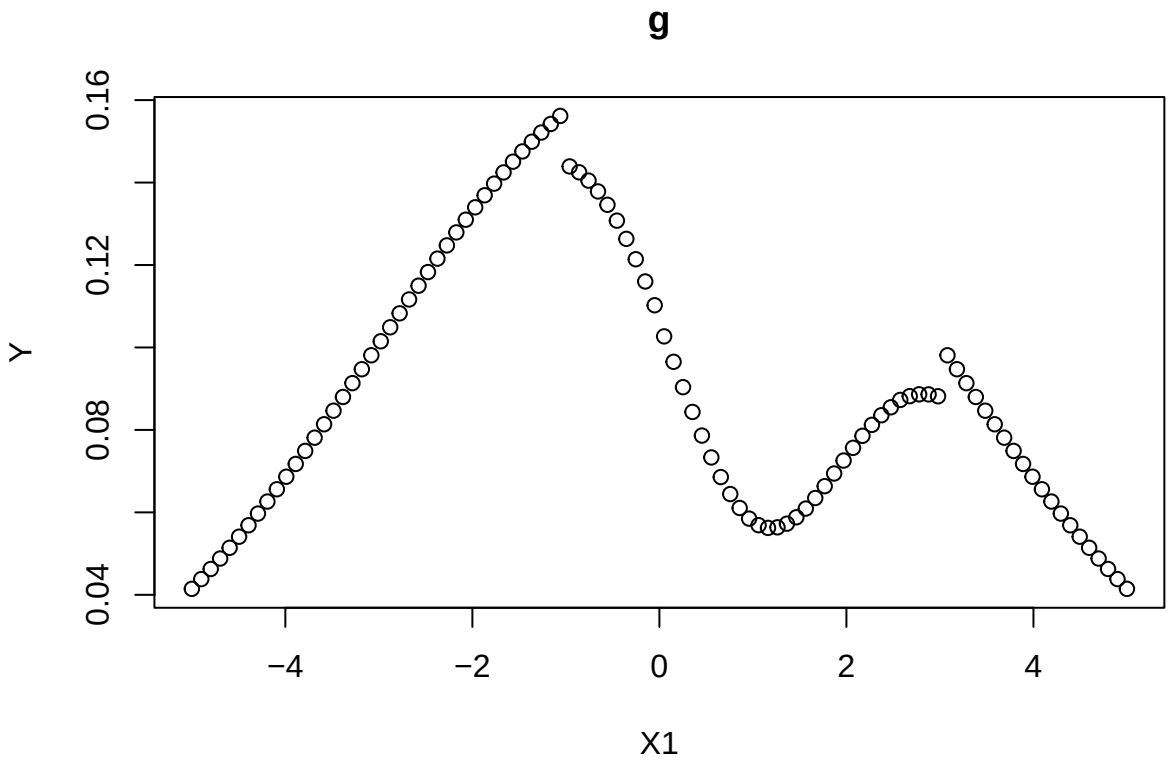



```r
#define g
g<-function(x,p){
```

```
  b = p[[1]]
  c = p[[2]]

  if (x <= b || x >= c){
    return (dnorm(x,mu_1,s_1)/Z)
  }else{
    i = findInterval(x,pp)
    sup = dnorm(inter_argmax(pp[i],pp[i+1],mu_1),mu_1,s_1)
    inf = dnorm(inter_argmin(pp[i],pp[i+1],mu_2),mu_2,s_2)
    return ((sup-a*inf)/Z)
  }
}

#graphically check that g is correct
X1=seq(-5,5,length.out=100)
Y=sapply(X1,function(i) g(i,P0))
plot(X1,Y,main="g")
```




```
#define a function that generates one sample X ~ f
accept_reject_once<-function(p){
  u = runif(1)
  y = F_h_inverse(p,runif(1)) #generates one sample Y ~ h using inverse function method
  i = 1
  while(u>f(a, mu_1, mu_2, s_1, s_2, y)/g(y,p)){
    u = runif(1)
    y = F_h_inverse(p,runif(1))
```

```
    i = i + 1
  }
  return (list(rv=y,nt=i))
}

#define a function generate n sample of f using accept_reject_once function n times
stratified<-function(n){
  X = numeric(n)
  s = 0
  for (j in (1:n)){
    res = accept_reject_once(P0)
    X[j] = res$rv
    s = s + res$nt
  }
  return (list(X_sam = X, accep_rate = n/s))
}
```

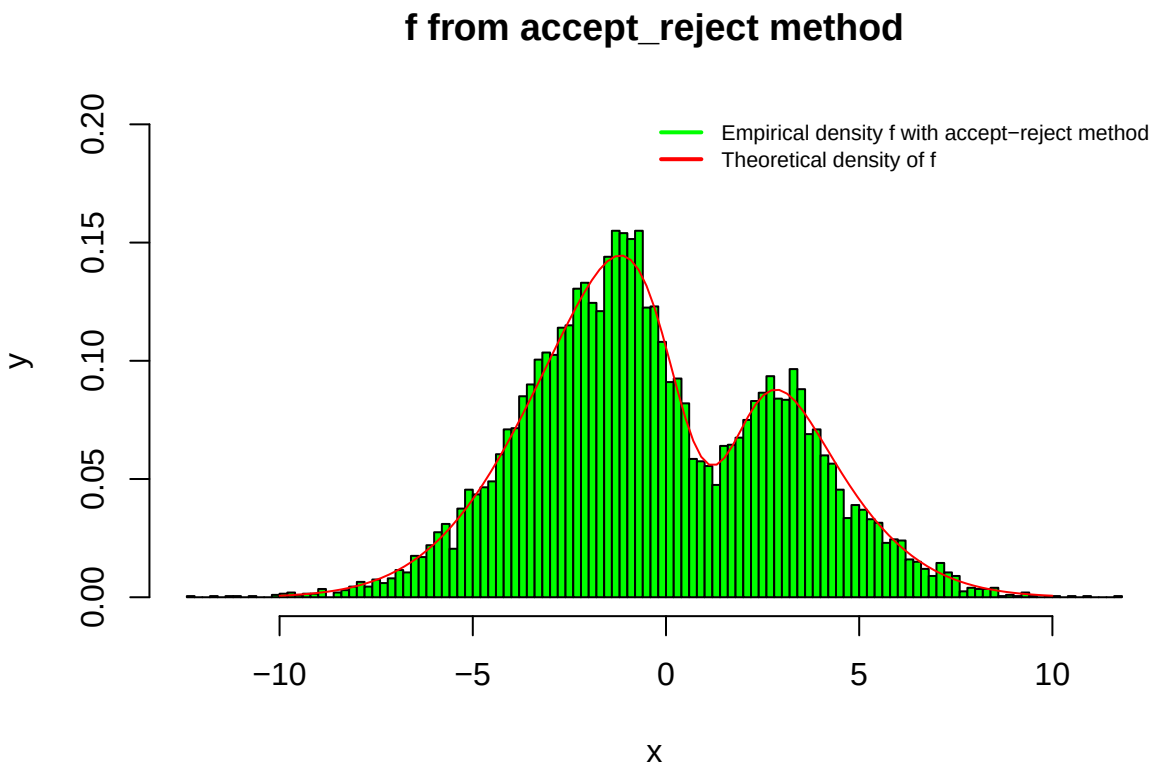
```
result = stratified(10000)

accep_rate = result$accep_rate
X = result$X_sam
theore_rate = 1/Ig
print(paste("Empirical acceptance rate:", accep_rate, "Theoretical rate",theore_rate))
```

Test

```
## [1] "Empirical acceptance rate: 0.986193293885602 Theoretical rate 0.98574512956691"
```

```
hist(X,freq=FALSE, col='green', breaks = 100, ylim = c(0,0.2),
     main = "f from accept_reject method", xlab="x",ylab="y")
x2<-seq(-10, 10,length.out = 100)
y2<-f(a, mu_1, mu_2, s_1, s_2, x2)
lines(x2,y2, col='red')
legend("topright", legend = c("Empirical density f with accept-reject method",
                              "Theoretical density of f"),
       col = c("green", "red"), lwd =2, cex= 0.7, bty="n")
```



The empirical acceptance rate of 0.986 is very close to the theoretical rate of 0.985 for a given partition:

$$D_0 =]-\infty, -1[\cup]3, +\infty[,$$

$$D_i =]-1+\frac{1}{40}(i-1), -1+\frac{1}{40}i], i = 1, ..., 160$$

Question 12: stratified(n,delta)

Constuct a Parition for a given delta Let $\delta < \frac{1}{Ig}$. We have $\left|Ig - 1\right| < \frac{1}{\delta} - 1\Big|$ $(*)$. We denote $\epsilon = \frac{2}{3}(\frac{1}{\delta} - 1)$. To achieve $(*)$, by error analysis, the D_0 of a Partition could be:

$$D_0 = \left]-\infty, q_{\epsilon/4}\right] \cup \left]q_{1-\epsilon/4}, +\infty\right[$$

where $q_{\epsilon/4}$ and $q_{1-\epsilon/4}$ is the quantile of $N(\mu_2, \sigma_2^2)$ of order $\epsilon/4$ and $1-\epsilon/4$. For the intervals D_i, $i = 1, .., k$, we consider equal lengths denoted as l to simplify the problem. In this case, by error analysis, l needs to satisfy two conditions:

$$l \leq \frac{\epsilon\sigma_1^2}{\max(|q_\epsilon - \mu_1|, |q_{1-\epsilon} - \mu_1|)}$$

and

$$l \leq \frac{\epsilon\sigma_2^2}{2\max(|q_\epsilon - \mu_2|, |q_{1-\epsilon} - \mu_2|)}$$

We take l as

$$\min\left(\frac{\epsilon\sigma_1^2}{\max(|q_\epsilon-\mu_1|,|q_{1-\epsilon}-\mu_1|)},\frac{\epsilon\sigma_2^2}{2\max(|q_\epsilon-\mu_2|,|q_{1-\epsilon}-\mu_2|)}\right)$$

```
#define a function to construct a partition for a given delta
Part_delta<-function(delta){
  eps = (1/delta-1)*2/3
  if(eps>1) {eps = 1} #ensure the input of qnorm is valid
  b = qnorm(eps/4,mu_2,s_2)
  c = qnorm(1-eps/4,mu_2,s_2)
  l = min((eps*s_1^2)/max(abs(b-mu_1),abs(c-mu_1)),(0.5*eps*s_2^2)/max(abs(b-mu_2),abs(c-mu_2)))

  return (P(b,c,ceiling((c-b)/l)))
}

#calculate Ig, pp, lh for a given partition constructed by delta
#call stratified(n)
oldstratified = stratified

stratified<-function(n,delta){
  P0 <<- Part_delta(delta)
  res_Ig = Integral_g(P0)
  Ig <<- res_Ig$Ig
  pp <<- res_Ig$pp
  lh <<- res_Ig$lh
  res = oldstratified(n)
  return (list(res = res,Par = P0,Ig=Ig))
}
```

```
delta = 0.9
result = stratified(5000,delta)
Ig_delta = result$Ig

PP = result$Par
print(paste("The Partition for delta =", delta, "is:",PP))
```

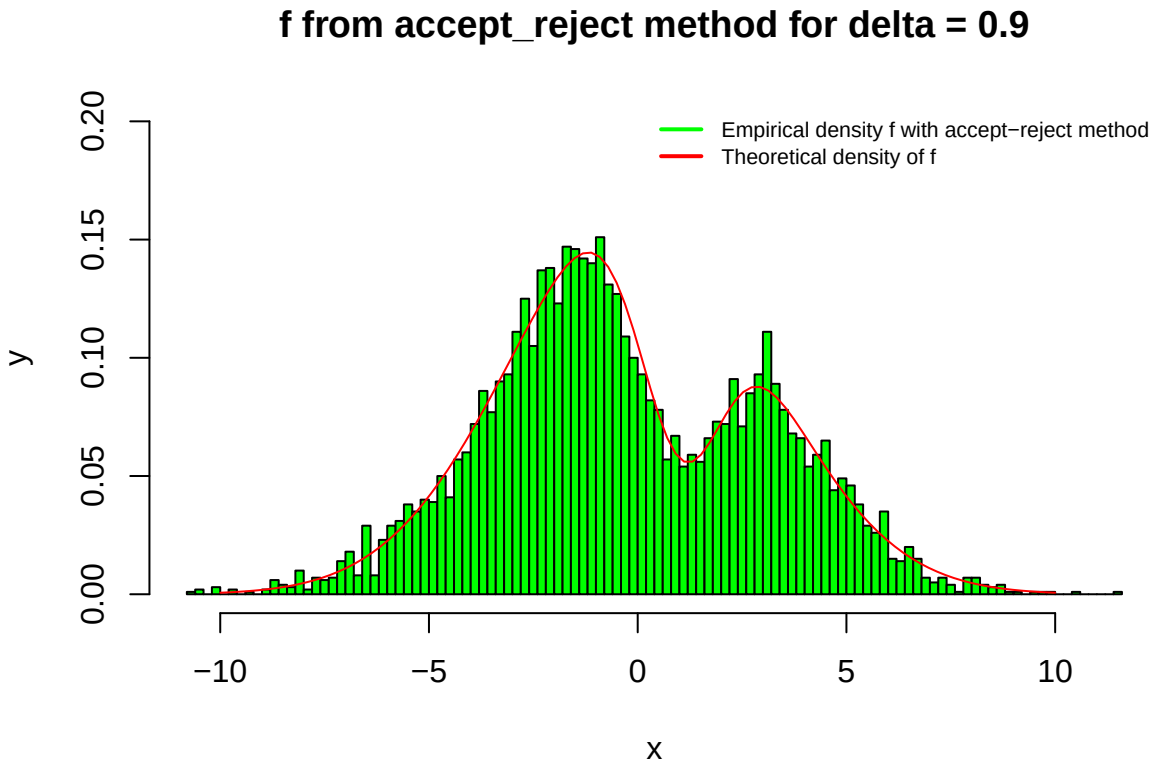
```
## [1] "The Partition for delta = 0.9 is: -1.08535556603183"
## [2] "The Partition for delta = 0.9 is: 3.08535556603183"
## [3] "The Partition for delta = 0.9 is: 235"
```

```
accep_rate_delta = result$res$accep_rate
X2 = result$res$X_sam
theore_rate_delta = 1/Ig_delta

print(paste("For a given delta =",delta,"the empirical acceptance rate is", accep_rate_delta, "and the
```

```
## [1] "For a given delta = 0.9 the empirical acceptance rate is 0.990295107942167 and the theoretical
```

```
hist(X2,freq=FALSE, col='green', breaks = 100, ylim = c(0,0.2),
     main = "f from accept_reject method for delta = 0.9", xlab="x",ylab="y")
x3<-seq(-10, 10,length.out = 100)
y3<-f(a, mu_1, mu_2, s_1, s_2, x2)
lines(x3,y3, col='red')
legend("topright", legend = c("Empirical density f with accept-reject method",
                              "Theoretical density of f"),
       col = c("green", "red"), lwd =2, cex= 0.7, bty="n")
```



The empirical acceptance rate is 0.985 which is bigger than the given $\delta = 0.9$.

The partition is:

$$D_0 =]-\infty, -1.085] \cup]3.085, \infty[, D_i =]-1.085 + 4/235 * (i-1), -1.085 + 4/235 * i], i = 1, ..., 235$$

Question 13: Monte Carlo Fn estimator

$$F_X(x) = P(X \leq x) = \mathbb{E}[1_{X \leq x}]$$

The Monte Carlo estimator is:

$$\frac{1}{n} \sum_{i=1}^{n} 1_{X_i \leq x}$$

Question 14: strong consistency of Fn(x)

$(1_{X_i \leq x})_{i=1}^n$ are i.i.d variables

$$\forall i \in \{1, \ldots, n\}, \quad 1_{X_i \leq x} \sim \text{Ber}(p), \text{ where } p = P(X \leq x) = F_X(x)$$

The law of large numbers gives that

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} 1_{X_i \leq x} \xrightarrow{\text{a.s.}}_{n \to \infty} \mathbb{E}[1_{X_i \leq x}] = F_X(x)$$

So F_n(x) is a strong consistent estimator of $F_X(x)$.

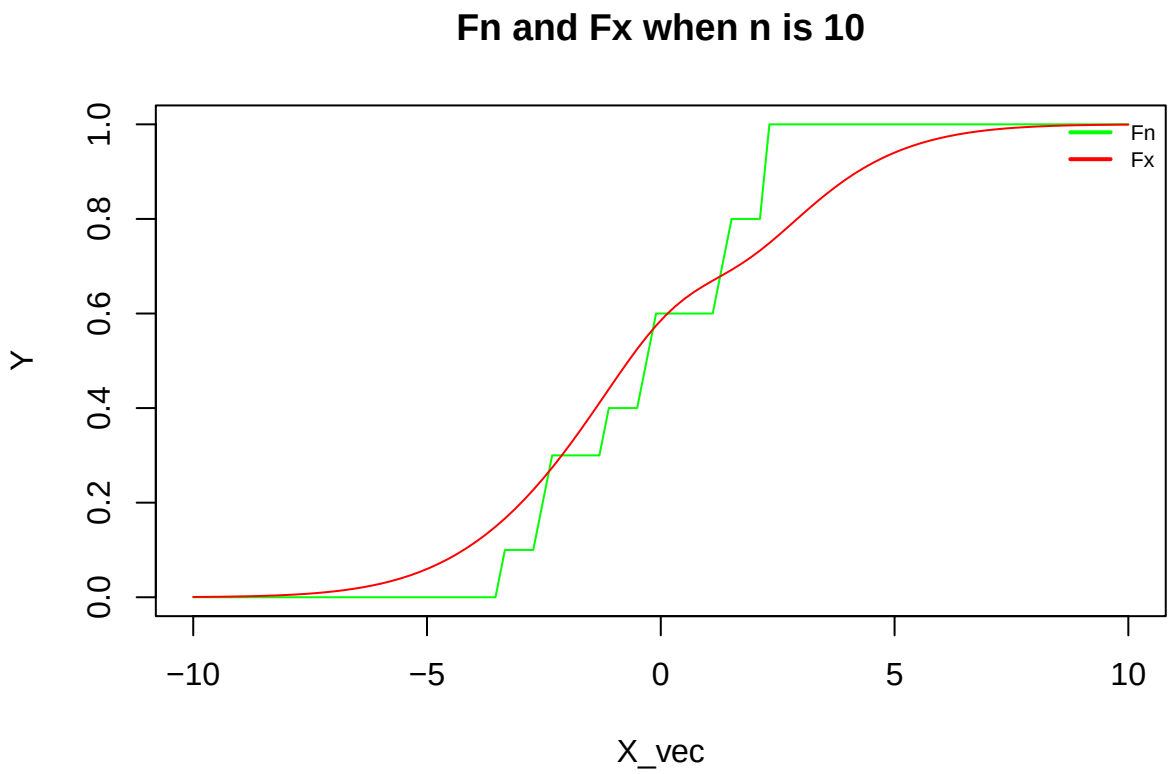
Question 15 : empirical cdf(x,Xn)

```
empirical_cdf<-function(x, Xn){ # Xn is a random vector of X1, ..., Xn
  count<-sum(Xn <= x)
  n = length(Xn)
  return (count/n)
}
```

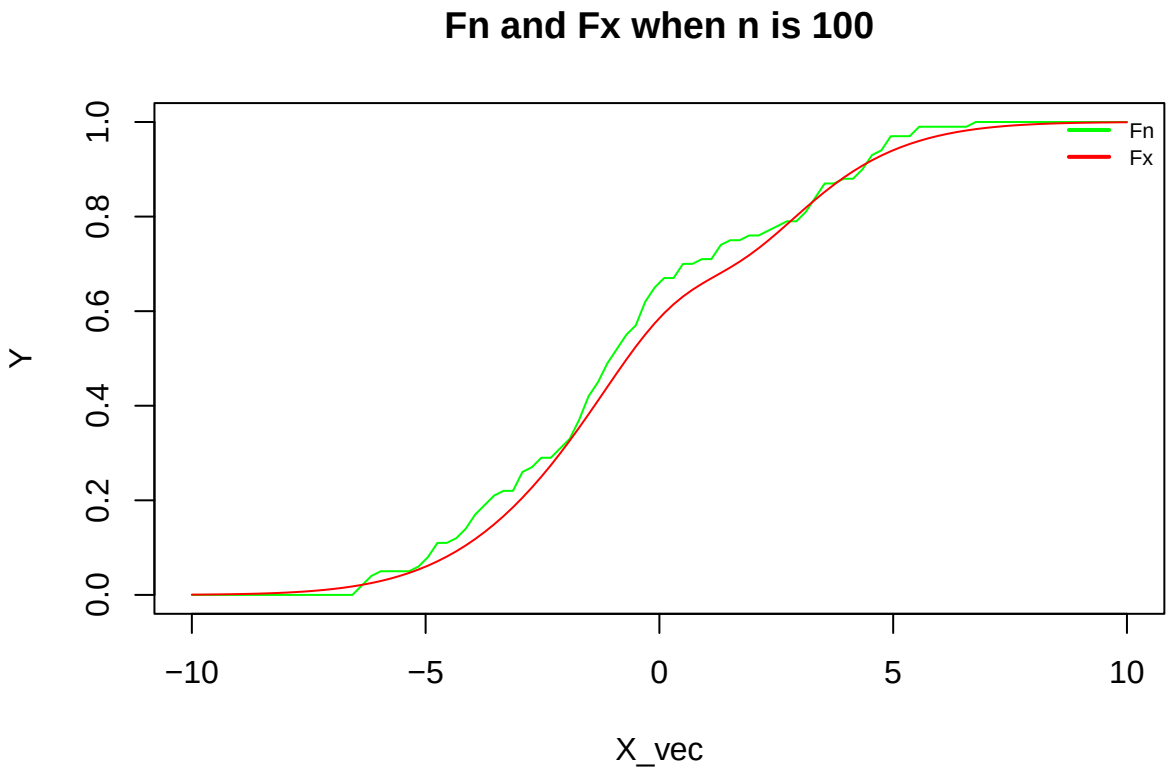
```
X_vec<-seq(-10,10,length.out=100)
N <- c(10, 100, 1000, 10000, 100000)
delta = 0.9

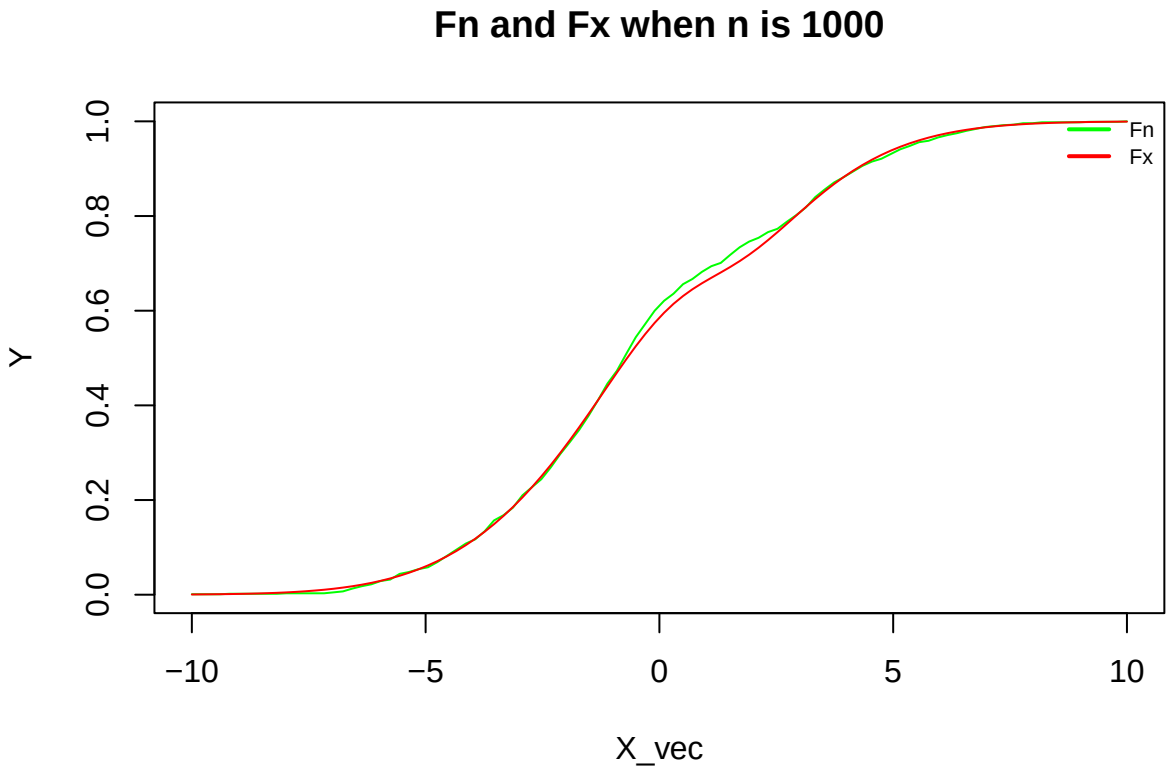
F_x = sapply(X_vec,function(i) F(a, mu_1, mu_2, s_1, s_2, i) )

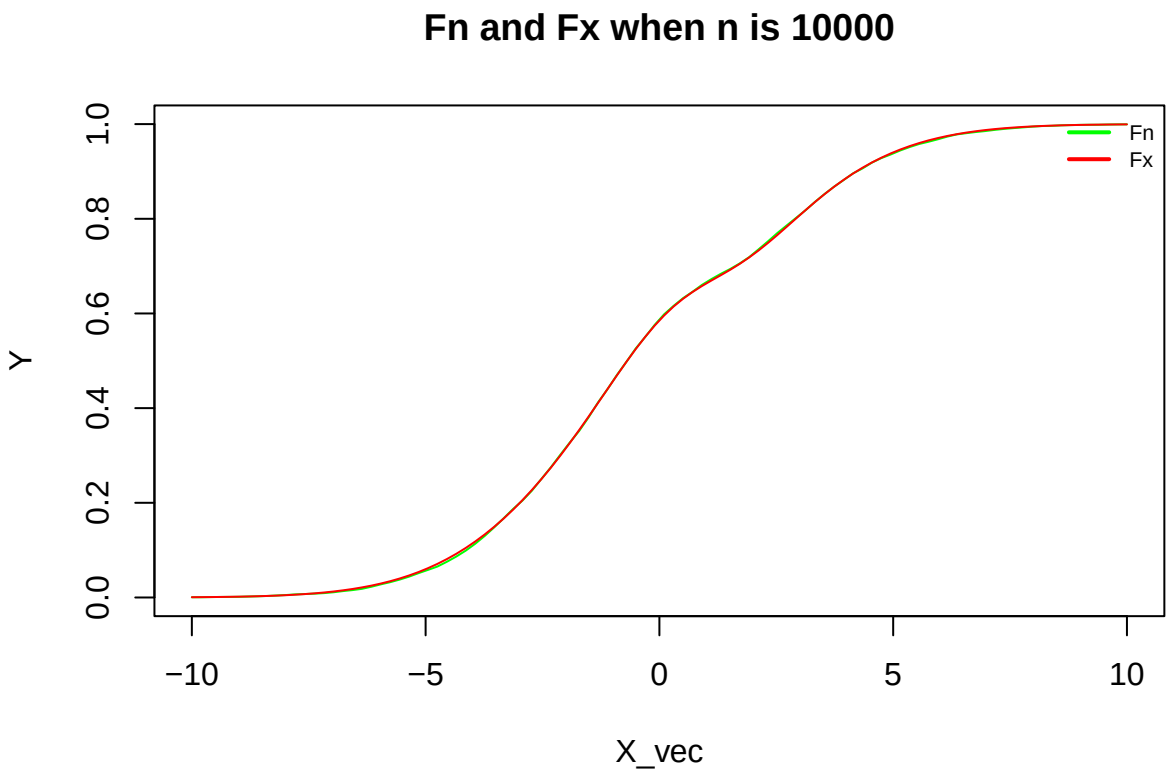
for (n in N){
  results = stratified(n,delta)
  X_n = results$res$X_sam
  Y = sapply(X_vec,function(i) empirical_cdf(i,X_n))
  #if (n == N[1]){
    plot(X_vec,Y,type="l",col="green", main=paste("Fn and Fx when n is",n))
    lines(X_vec,F_x,col="red")
    legend("topright", legend = c("Fn","Fx"), col = c("green", "red"), lwd =2, cex= 0.7, bty="n")
}
```

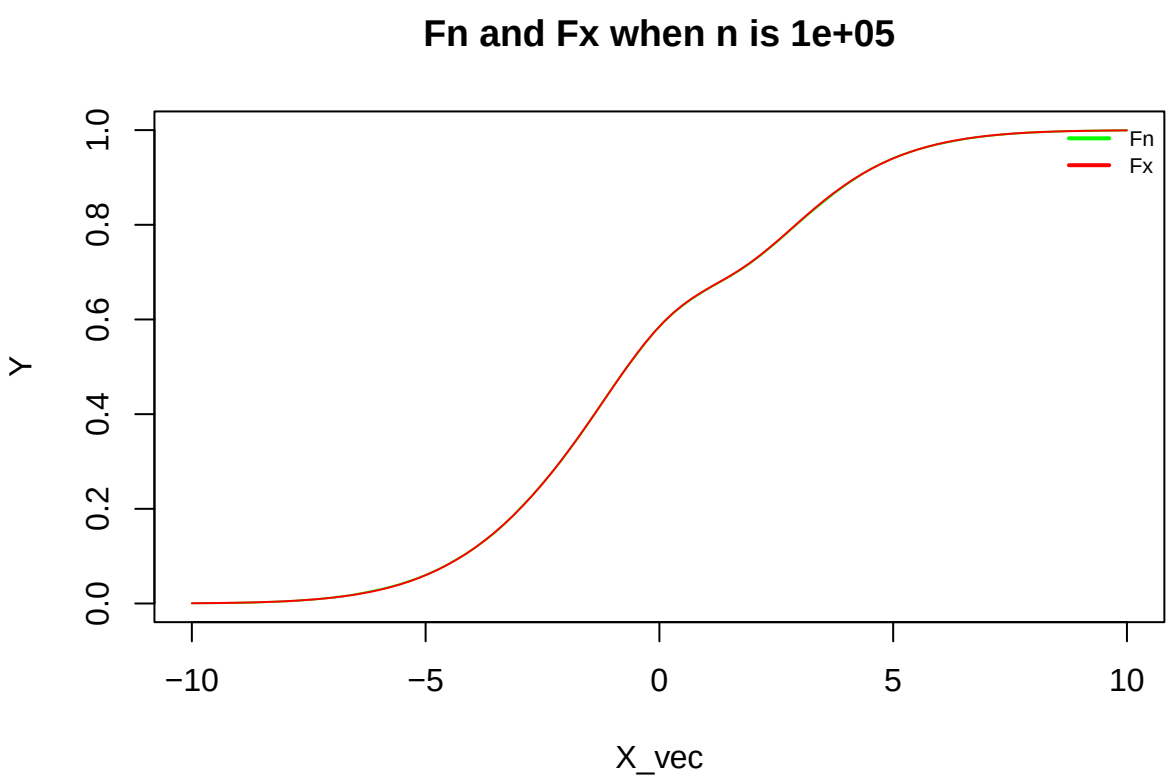

Fn and Fx when n is 10
Fn
Fx
Y
0.0
0.2
0.4
0.6
0.8
1.0
−10
−5
0
5
10
X_vec

test


Fn and Fx when n is 100
Fn
Fx
Y
0.0
0.2
0.4
0.6
0.8
1.0
−10
−5
0
5
10
X_vec


Fn and Fx when n is 1000
Y
0.0
0.2
0.4
0.6
0.8
1.0
−10
−5
0
5
10
X_vec
Fn
Fx


Fn and Fx when n is 10000
Y
0.0
0.2
0.4
0.6
0.8
1.0
−10
−5
0
5
10
X_vec
Fn
Fx



From these grapes, we can observe the strong consistency of F_n when n is increasing.

Question 16

Let $(Y_i)_{i=1}^n$ be n i.i.d square integrable random variables.

The central limit theorem states that:

$$\sqrt{n}\left(\overline{Y_n} - \mathbb{E}[Y_i]\right) \xrightarrow{d}_{n\to\infty} \mathcal{N}(0, \mathrm{Var}(Y_1))$$

Here we apply the CLT to $Y_i = 1_{X_i \leq x}$.

We know that $1_{X_i \leq x} \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$, where $p = F_X(x)$.

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} 1_{X_i \leq x} - \mathbb{E}[1_{X_i \leq x}]\right) \xrightarrow{D}_{n\to\infty} \mathcal{N}(0, \mathrm{Var}(1_{X_i \leq x}))$$

$$\Rightarrow \quad \sqrt{n}\left(F_n(x) - F_X(x)\right) \xrightarrow{D}_{n\to\infty} \mathcal{N}(0, p(1-p))$$

$$\Rightarrow \quad \sqrt{n}\frac{F_n(x) - F_X(x)}{\sqrt{p(1-p)}} \xrightarrow{D}_{n\to\infty} \mathcal{N}(0,1)$$

$$\mathrm{CI}_{95\%} = \left[F_n(x) \pm \frac{q_{0.975} \cdot \sqrt{F_n(x)(1-F_n(x))}}{\sqrt{n}}\right]$$

where $q_{0.975}$ is the quantile of $N(0,1)$ of order 0.975.

Question 17: 95% CI of F(x) for a given x

```
q<- qnorm(0.975)

#we denote eps the absolute precision of Fn(x)
simu_IC<-function(x,eps_rel){
  Fx <-F(a, mu_1, mu_2, s_1, s_2, x)
  eps = eps_rel*Fx
  return (ceiling((Fx-Fx^2)/((eps^2)/(q^2))))
}
```

```
eps_rel = 0.01
print(paste("When x = 1, the number of simulation needed to have a 95% IC is", simu_IC(1,eps_rel)))
```

test

```
## [1] "When x = 1, the number of simulation needed to have a 95% IC is 19509"
```

```
print(paste("When x = -15, the number of simulation needed to have a 95% IC is", simu_IC(-15,eps_rel)))
```

```
## [1] "When x = -15, the number of simulation needed to have a 95% IC is 107209108789"
```

We observe that: At $x = 1$, the CDF $F_X(x)$ is moderate, so n is reasonable.
At $x = -15$, the CDF $F_X(x)$ is extremely small, so n becomes extremely large. This reflects the difficulty of estimating extreme tail probabilities accurately with the Monte Carlo method.

Question 18: Qn(u)

Let $u \in (0, 1)$. Let $n \in \mathbb{N}$. We have:

$$u \leq F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{X_i \leq x}.$$

So, we can express:

$$un \leq \sum_{i=1}^{n} \mathbb{1}_{X_i \leq x}.$$

We order $(X_i)_{i=1}^n$ in ascending order to obtain a new sequence:

$$(X_{(1)}, X_{(2)}, \ldots, X_{(n)}).$$

The quantile function $Q_n(u)$ is defined as:

$$Q_n(u) = \begin{cases} X_{(un)}, & \text{if } un \text{ is an integer,} \\ X_{\lceil un \rceil}, & \text{otherwise.} \end{cases}$$

Where:

$$\lceil un \rceil := \min\{k \in \mathbb{Z} \mid k \geq un\}.$$

Question 19: TCL for Qn(u)

We have:

$$\sum_{j=1}^{n} Y_{j,n} = nF_n(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}).$$

Then we have: - Expectation:

$$\mathbb{E}[Y_{j,n}] = P\left(X_j \leq Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right) = F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right).$$

- Variance:

$$\mathrm{Var}(Y_{j,n}) = F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right)\left(1 - F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right)\right).$$

- S_n:

$$s_n = n\sum_{j=1}^{n}\sigma_{j,n}^2 = \sqrt{n}\mathrm{Var}(Y_{j,n}).$$

We have:

$$\frac{\sum_{j=1}^{n} Y_{j,n} - n\mathbb{E}[Y_{j,n}]}{s_n} = \frac{\sqrt{n}\left(F_n(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}) - F_X(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))})\right)}{\sqrt{F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right)\left(1 - F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right)\right)}}.$$

Let:

$$\sigma_n = \frac{\sqrt{u(1-u)}}{f(Q(u))}.$$

We have:

$$F_X\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right) = F_X(Q(u)) + f_X(Q(u))\cdot t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))} + o\left(\frac{1}{n}\right), \quad \text{(applying Taylor development)}.$$

This simplifies to:

$$= u + t\frac{\sqrt{u(1-u)}}{\sqrt{n}} + o\left(\frac{1}{n}\right).$$

We also have:

$$\frac{s_n}{n} = \sqrt{\frac{n(u - u^2 + O(1/\sqrt{n}))}{n^2}} = \sqrt{\frac{u-u^2}{n} + O\left(\frac{1}{n^{3/2}}\right)}.$$

Thus, it can be rewritten as:

$$\frac{\sum_{j=1}^{n} Y_{j,n} - n\mathbb{E}[Y_{j,n}]}{S_n} = \frac{\sum_{j=1}^{n} Y_{j,n} - \mathbb{E}[Y_{j,n}]}{\frac{S_n}{n}}.$$

Substituting:

$$= \frac{F_n\left(Q(u) + t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}\right) - \left(u + t\frac{\sqrt{u(1-u)}}{\sqrt{n}} + O\left(\frac{1}{\sqrt{n}}\right)\right)}{\sqrt{\frac{u-u^2}{n} + O\left(\frac{1}{n^{3/2}}\right)}}.$$

For the numerator:

$$F_n(Q(u)+t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))})-u-t\sqrt{\frac{u(1-u)}{n}+O\left(\frac{1}{n}\right)}>-t\sqrt{\frac{u(1-u)}{n}+O\left(\frac{1}{n^{\frac{3}{2}}}\right)}.$$

Thus:

$$F_n(Q(u)+t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))})>u+O\left(\frac{1}{\sqrt{n}}\right).$$

We also have:

$$Q_n(F_n(Q(u)+t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}))>Q_n(u)+O\left(\frac{1}{\sqrt{n}}\right).$$

Expanding further:

$$Q(u)+t\frac{\sqrt{u(1-u)}}{\sqrt{n}f(Q(u))}>Q_n(u)+O\left(\frac{1}{\sqrt{n}}\right).$$

Thus, we have:

$$P\left(\sqrt{n}\frac{Q_n(u)-Q(u)}{f(Q(u))/\sqrt{u(1-u)}}<t+O\left(\frac{1}{\sqrt{n}}\right)\right)\xrightarrow{n\to\infty}\Phi(t),$$

where Φ is the CDF of $N(0,1)$.

Finally, we get:

$$P\left(\sqrt{n}\frac{Q_n(u)-Q(u)}{f(Q(u))/\sqrt{u(1-u)}}<t\right)\xrightarrow{n\to\infty}\Phi(t),$$

which terminates the proof.

Question 20 Behaviour of u -> 0 or 1

When $u\to 0$ or 1, $Q(u)\to\infty$.

We have:

$$\frac{f(Q(u))}{\sqrt{u(1-u)}}\xrightarrow{u\to 0 \text{ or } 1}0.$$

In fact, we have:

$$\frac{f(Q(u))}{\sqrt{u(1-u)}}=\frac{1}{Q'(u)\sqrt{u(1-u)}},$$

because:

By differentiating $F_X(Q(u))=u$, we get:

$$f_X(Q(u))Q'(u)=1.$$

So we obtain:

$$f_X(Q(u))=\frac{1}{Q'(u)}.$$

Thus, we rewrite it as:

$$\frac{1}{\sqrt{u(1-u)}Q'(u)}.$$

Since the numerator $\frac{1}{\sqrt{u(1-u)}}$ is integrable when $u\to 0$ or 1, while $Q'(u)$ isn't integrable when $u\to 0$ or 1, we conclude:

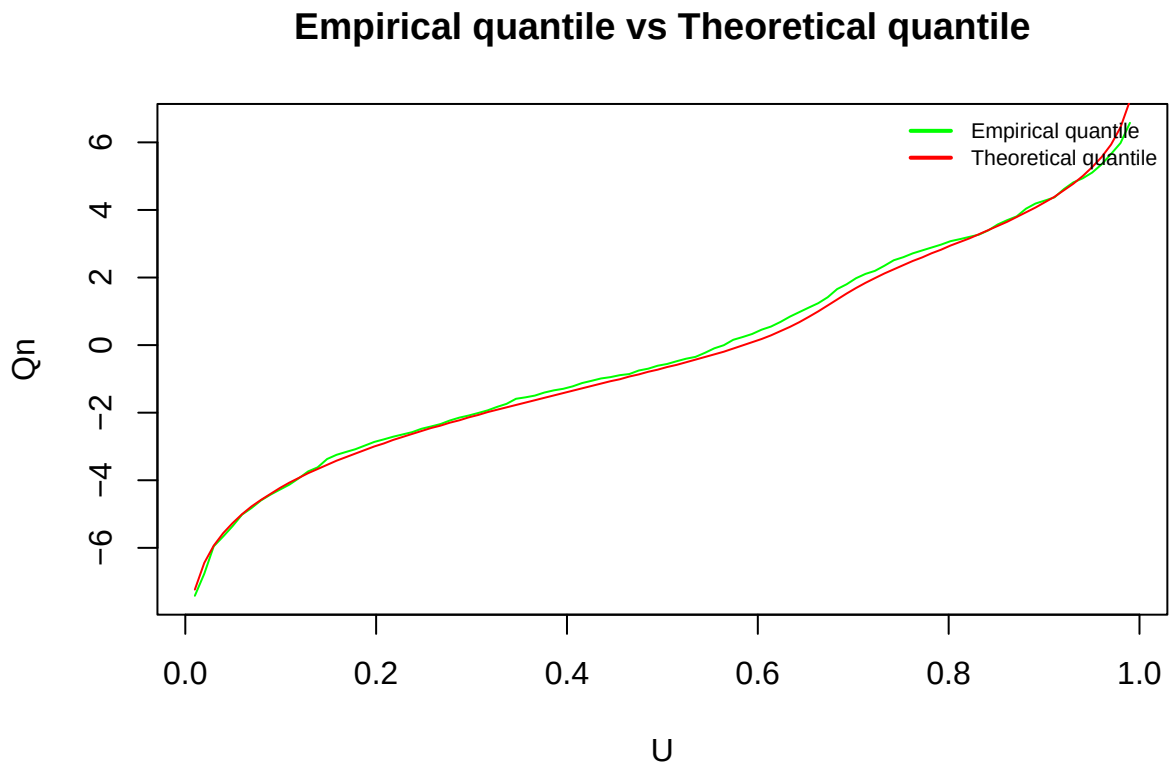
$$\frac{1}{\sqrt{u(1-u)}Q'(u)}\xrightarrow{u\to 0 \text{ or } 1}0.$$

Question 21

```
empirical_quantile<-function(u,Xn){
  n <- length(Xn)
  sorted_Xn<-sort(Xn)
  ind<- ceiling(u*n)
  return(sorted_Xn[ind])
}

delta = 0.9
n =1000
result1 = stratified(n,delta)
Xn = result1$res$X_sam
U<-seq(0.01,0.99, length.out = 100)
Qn<-sapply(U, function(u) empirical_quantile(u,Xn))

plot(U,Qn, type="l", main="Empirical quantile vs Theoretical quantile",col="green")
Q = sapply(U,function(u) inverse_F(u,a, mu_1, mu_2, s_1, s_2,eps=0.01))
lines(U,Q,col="red")
legend("topright", legend = c("Empirical quantile","Theoretical quantile"), col = c("green", "red"), lwd
```



Question 22

```
#we denote eps the absolute precision of Qn(x)
IC_quantile<-function(u,eps_rel){
```

```
  Q_u<-inverse_F(u,a, mu_1, mu_2, s_1, s_2,eps_rel*0.5)
  eps = abs(Q_u*eps_rel)
  #print(u)
  #print(Q_u*eps_rel)
  Q_u<-inverse_F(u,a, mu_1, mu_2, s_1, s_2,eps/4)
  #print(Q_u)
  eps = abs(Q_u*eps_rel*3/4)

  f_Q_u<-f(a,mu_1,mu_2,s_1,s_2,Q_u)
  #print(u)
  return(ceiling((q*q*f_Q_u*f_Q_u)/(eps^2*(u*(1-u)))))
}

U<-c(0.5,0.9, 0.99, 0.999, 0.9999)
result<-sapply(U, function(u) IC_quantile(u,eps_rel =0.01))
print(result)
```

```
## [1] 10911   162    11     2     1
```